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(54) **TRANSISTOR AND MEMORY CIRCUITRY
COMPRISING STRINGS OF MEMORY
CELLS**

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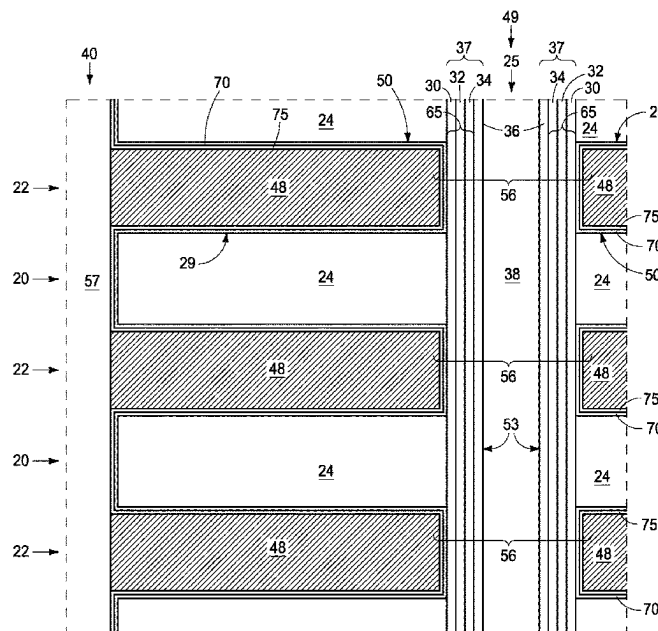
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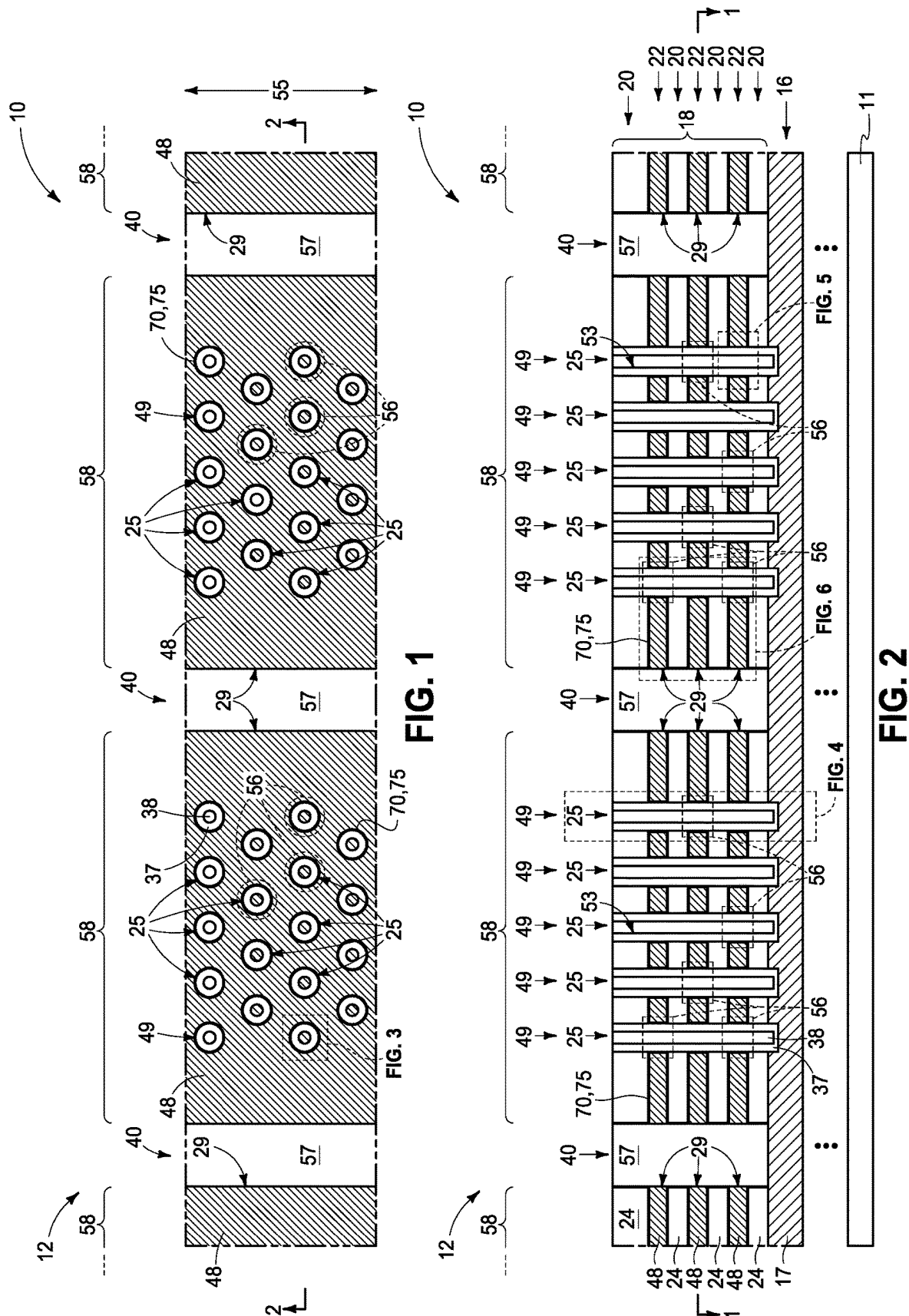
(57) **ABSTRACT**

Memory circuitry comprising strings of memory cells comprises a vertical stack comprising alternating insulative tiers and conductive tiers. Channel-material strings extend through the insulative tiers and the conductive tiers. Charge-passage material is in the conductive tiers laterally-outward of the channel-material strings. Storage material is in the conductive tiers laterally-outward of the charge-passage material. At least one of AlO_x, ZrO_x, and HfO_x is in the conductive tiers laterally-outward of the storage material. At least one of (a) and (b) is in the conductive tiers laterally-outward of the at least one of AlO_x, ZrO_x, and HfO_x, where, (a): MoO_xN_y, where each of “x” and “y” is from 0 to 4.0; and (b): MoM_z, where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0. Metal material is in the conductive tiers laterally-outward of the at least one of the (a) and the (b). Memory cells are in individual of the conductive tiers. The memory cells individually comprise the channel material of individual of the channel-material strings, the storage material, the at least one of AlO_x, ZrO_x, and HfO_x, the at least one of the (a) and the (b), and the metal material. Other embodiments are disclosed.

20 Claims, 4 Drawing Sheets



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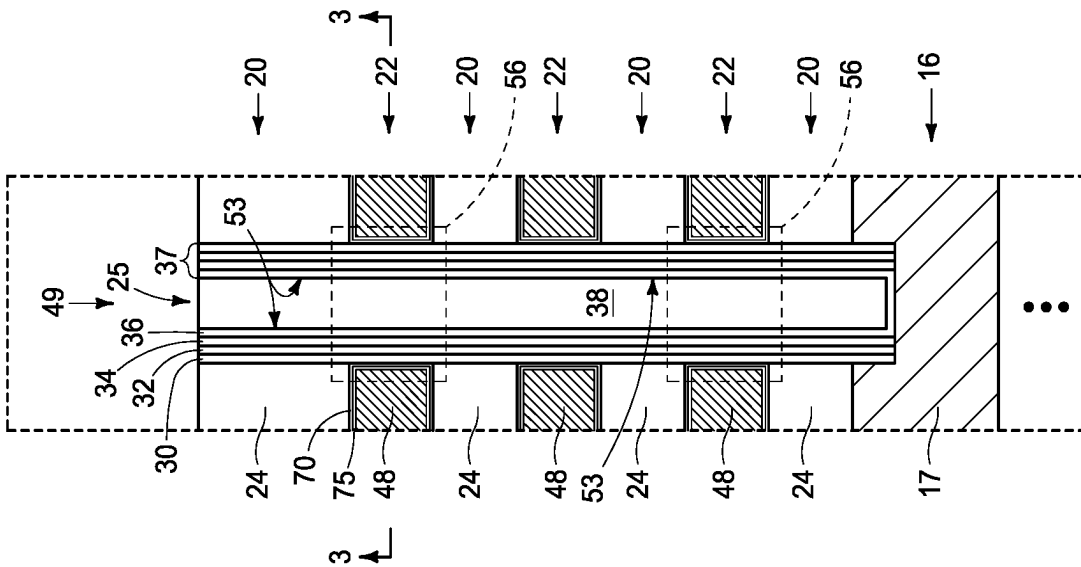


FIG. 4

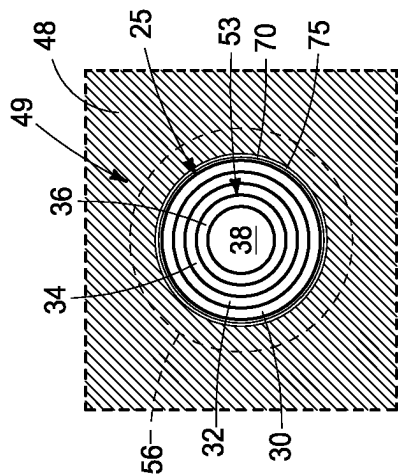
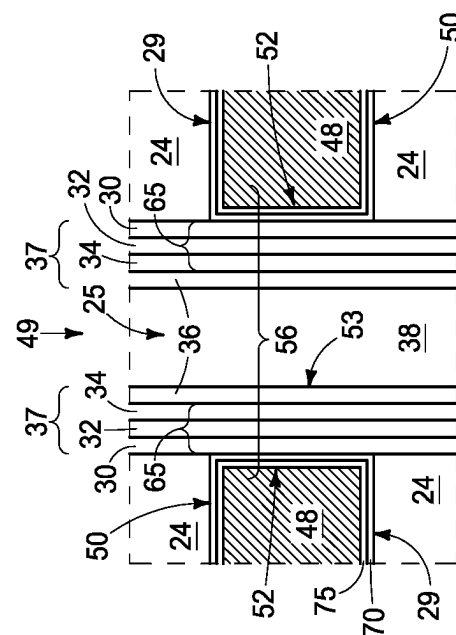
**FIG. 3**

FIG. 5

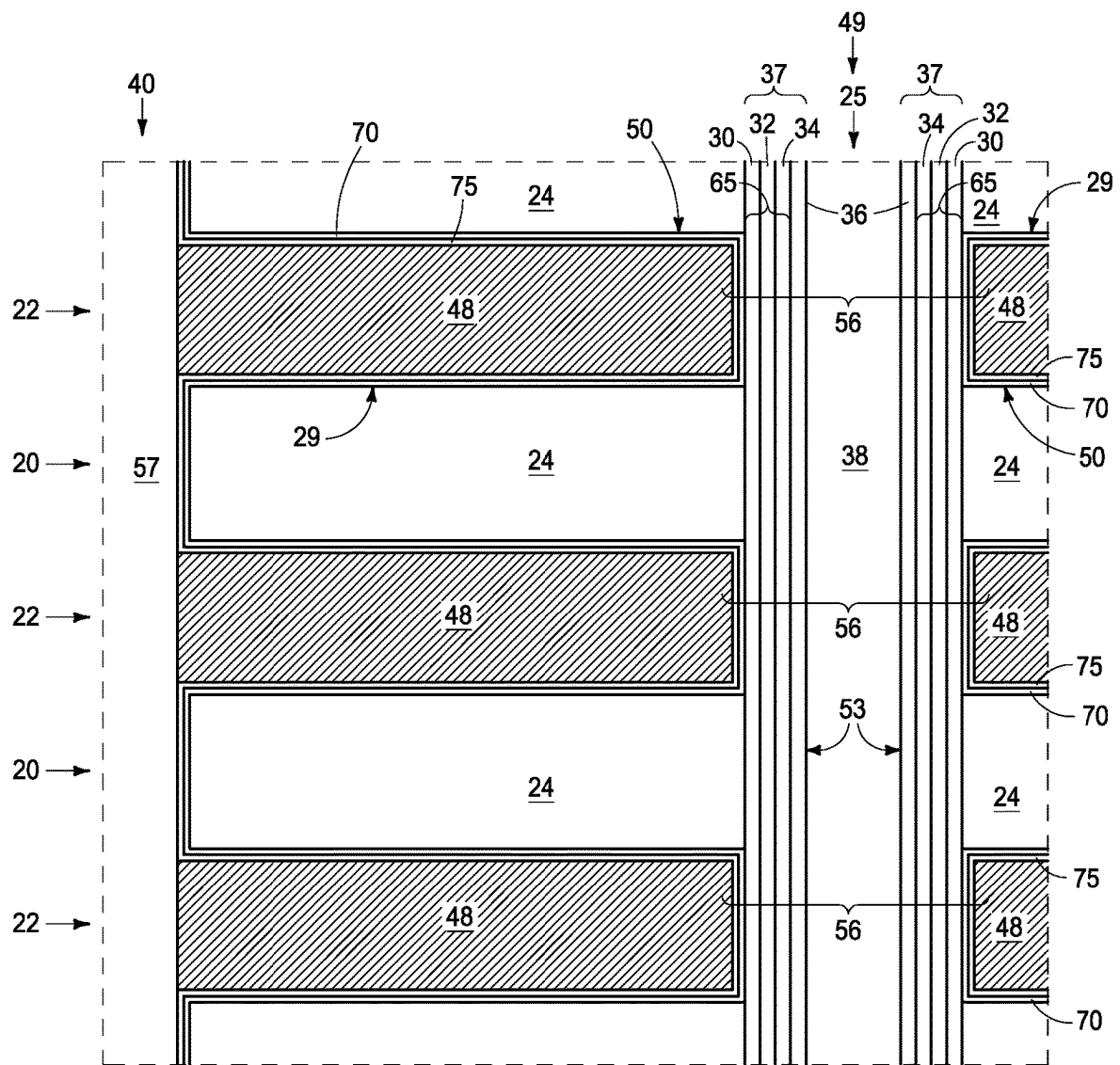


FIG. 6

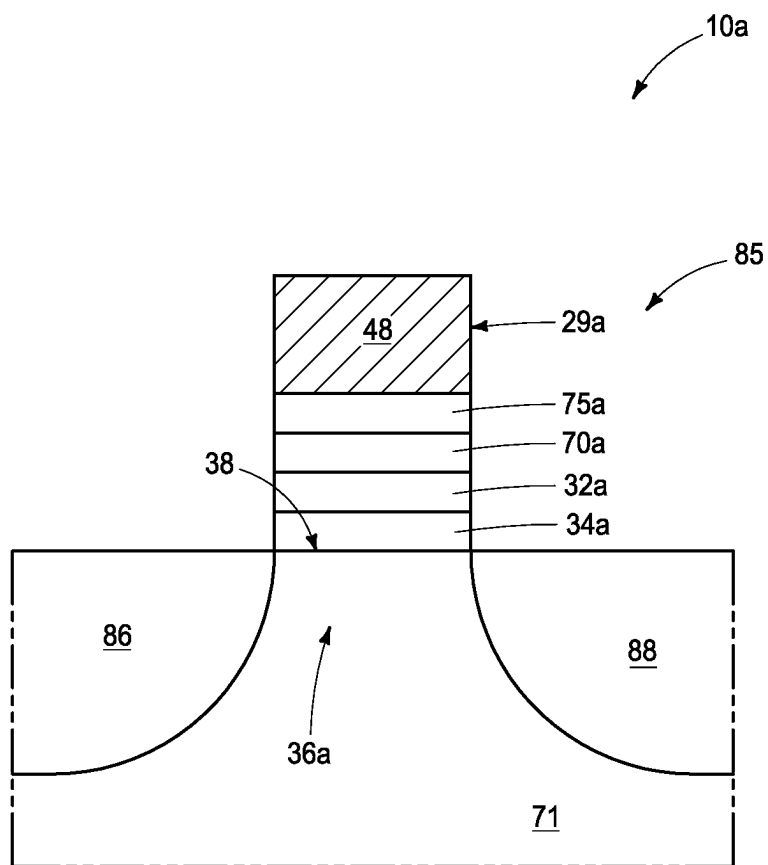


FIG. 7

1

TRANSISTOR AND MEMORY CIRCUITRY COMPRISING STRINGS OF MEMORY CELLS

TECHNICAL FIELD

Embodiments disclosed herein pertain to transistors and to memory circuitry comprising strings of memory cells.

BACKGROUND

Memory is one type of integrated circuitry and is used in computer systems for storing data. Memory may be fabricated in one or more arrays of individual memory cells. Memory cells may be written to, or read from, using digitlines (which may also be referred to as bitlines, data lines, or sense lines) and access lines (which may also be referred to as wordlines). The sense lines may conductively interconnect memory cells along columns of the array, and the access lines may conductively interconnect memory cells along rows of the array. Each memory cell may be uniquely addressed through the combination of a sense line and an access line.

Memory cells may be volatile, semi-volatile, or non-volatile. Non-volatile memory cells can store data for extended periods of time in the absence of power. Non-volatile memory is conventionally specified to be memory having a retention time of at least about 10 years. Volatile memory dissipates and is therefore refreshed/rewritten to maintain data storage. Volatile memory may have a retention time of milliseconds or less. Regardless, memory cells are configured to retain or store memory in at least two different selectable states. In a binary system, the states are considered as either a "0" or a "1". In other systems, at least some individual memory cells may be configured to store more than two levels or states of information.

A field effect transistor is one type of electronic component that may be used in a memory cell. These transistors comprise a pair of conductive source/drain regions having a semiconductive channel region there-between. A conductive gate is adjacent the channel region and separated there-from by a thin gate insulator. Application of a suitable voltage to the gate allows current to flow from one of the source/drain regions to the other through the channel region. When the voltage is removed from the gate, current is largely prevented from flowing through the channel region. Field effect transistors may also include additional structure, for example a reversibly programmable charge-storage region as part of the gate construction between the gate insulator and the conductive gate.

Flash memory is one type of memory and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.

NAND may be a basic architecture of integrated flash memory. A NAND cell unit comprises at least one selecting device coupled in series to a serial combination of memory cells (with the serial combination commonly being referred to as a NAND string). NAND architecture may be configured in a three-dimensional arrangement comprising verti-

2

cally-stacked memory cells individually comprising a reversibly programmable vertical transistor. Control or other circuitry may be formed below the vertically-stacked memory cells. Other volatile or non-volatile memory array architectures may also comprise vertically-stacked memory cells that individually comprise a transistor.

Memory arrays may be arranged in memory pages, memory blocks and partial blocks (e.g., sub-blocks), and memory planes, for example as shown and described in any of U.S. Patent Application Publication Nos. 2015/0228651, 2016/0267984, and 2017/0140833. The memory blocks may at least in part define longitudinal outlines of individual wordlines in individual wordline tiers of vertically-stacked memory cells. Connections to these wordlines may occur in a so-called "stair-step structure" at an end or edge of an array of the vertically-stacked memory cells. The stair-step structure includes individual "stairs" (alternately termed "steps" or "stair-steps") that define contact regions of the individual wordlines upon which elevationally-extending conductive vias contact to provide electrical access to the wordlines.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagrammatic cross-sectional view of a portion of memory circuitry comprising strings of memory cells in accordance with an embodiment of the invention.

FIG. 2 is a diagrammatic cross-sectional view taken through line 2-2 in FIG. 1.

FIGS. 3-6 are enlarged views of portions of FIGS. 1 and 2.

FIG. 7 shows an alternate embodiment in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

Embodiments of the invention encompass memory circuitry, for example an array of NAND or other memory cells that may have at least some peripheral control circuitry under the array (e.g., CMOS-under-array). Embodiments of the invention also encompass a transistor regardless of whether part of a memory cell. First example embodiments are described with reference to FIGS. 1-6.

FIGS. 1-6 show an example construction 10 having an array 12 in which elevationally-extending strings 49 of transistors and/or memory cells 56 have been formed. Such includes a base substrate 11 having any one or more of conductive/conductor/conducting, semiconductive/semiconductor/semiconducting, or insulative/insulator/insulating (i.e., electrically herein) materials. Various materials have been formed elevationally over base substrate 11. Materials may be aside, elevationally inward, or elevationally outward of the FIGS. 1-6-depicted materials. For example, other partially or wholly fabricated components of integrated circuitry may be provided somewhere above, about, or within base substrate 11. Control and/or other peripheral circuitry for operating components within an array (e.g., array 12) of elevationally-extending strings of memory cells may also be fabricated and may or may not be wholly or partially within an array or sub-array. Further, multiple sub-arrays may also be fabricated and operated independently, in tandem, or otherwise relative one another. In this document, a "sub-array" may also be considered as an array.

A conductor tier 16 comprising conductor material 17 is above substrate 11. Conductor tier 16 may comprise part of control circuitry (e.g., peripheral-under-array circuitry and/or a common source line or plate) used to control read and

write access to the transistors and/or memory cells that will be formed within array 12. A vertical stack 18 comprising vertically-alternating insulative tiers 20 and conductive tiers 22 is above conductor tier 16. In some embodiments, conductive tiers 22 are referred to as first tiers 22 and insulative tiers 20 are referred to as second tiers 20. Example thickness for each of tiers 20 and 22 is 22 to 60 nanometers. The example uppermost tier 20 may be thicker/thickest compared to one or more other tiers 20 and/or 22. Only a small number of tiers 20 and 22 is shown, with more likely stack 18 comprising dozens, a hundred or more, etc. of tiers 20 and 22. Other circuitry that may or may not be part of peripheral and/or control circuitry may be between conductor tier 16 and stack 18. For example, multiple vertically-alternating tiers of conductive material and insulative material of such circuitry may be below a lowest of the conductive tiers 22 and/or above an uppermost of the conductive tiers 22. For example, one or more select gate tiers (not shown) may be between conductor tier 16 and the lowest conductive tier 22 and one or more select gate tiers may be above an uppermost of conductive tiers 22 (not shown). Alternately or additionally, at least one of the depicted uppermost and lowest conductive tiers 22 may be a select gate tier. Example insulative tiers 20 comprise insulative material 24 (e.g., silicon dioxide and/or other material that may be of one or more composition(s)).

Channel openings 25 have been formed (e.g., by etching) through insulative tiers 20 and conductive tiers 22 to conductor tier 16. Channel openings 25 may taper radially-inward (not shown) moving deeper in stack 18. In some embodiments, channel openings 25 may go into conductor material 17 of conductor tier 16 as shown or may stop there-atop (not shown). Alternately, as an example, channel openings 25 may stop atop or within the lowest insulative tier 20. A reason for extending channel openings 25 at least to conductor material 17 of conductor tier 16 is to assure direct electrical coupling of channel material to conductor tier 16 without using alternative processing and structure to do so when such a connection is desired. Etch-stop material (not shown) may be within or atop conductor material 17 of conductor tier 16 to facilitate stopping of the etching of channel openings 25 relative to conductor tier 16 when such is desired. Such etch-stop material may be sacrificial or non-sacrificial. By way of example and for brevity only, channel openings 25 are shown as being arranged in groups or columns of staggered rows of four and five openings 25 per row and being arrayed in laterally-spaced memory blocks 58. In this document, "block" is generic to include "sub-block". Memory blocks 58 may be considered as being longitudinally elongated and oriented, for example along a direction 55. Any alternate existing or future-developed arrangement and construction may be used.

Example memory blocks 58 are shown as at least in part having been defined by horizontally-elongated trenches 40 that were formed (e.g., by anisotropic etching) into stack 18. Trenches 40 will typically be wider than lower channel openings 25 (e.g., 10 to 20 times wider, yet such wider degree not being shown in FIGS. 1 and 2 for brevity). Trenches 40 may have respective bottoms that are directly against conductor material 17 (e.g., atop or within) of conductor tier 16 (as shown) or may have respective bottoms that are above conductor material 17 of conductor tier 16 (not shown). Intervening material 57 is in trenches 40 in stack 18 and may provide lateral electrical isolation (insulation) between immediately-laterally-adjacent memory blocks 58. Such may include one or more of insulative, semiconductive, and conducting materials and, regardless,

may facilitate conductive tiers 22 from shorting relative one another in a finished circuitry construction. Example insulative materials are one or more of SiO_2 , Si_3N_4 , Al_2O_3 , and undoped polysilicon. Intervening material 57 may include through array vias (not shown).

Transistor channel material may be formed in the individual channel openings elevationally along the insulative tiers and the conductive tiers, thus comprising individual channel-material strings, which is directly electrically coupled with conductive material in the conductor tier. Individual memory cells of the example memory array being formed may comprise a gate region (e.g., a control-gate region) and a memory structure laterally between the gate region and the channel material. In one such embodiment, the memory structure is formed to comprise a charge-blocking region, storage material (e.g., charge-storage material), and an insulative charge-passage material. The storage material (e.g., floating gate material such as doped or undoped silicon or charge-trapping material such as silicon nitride, metal dots, etc.) of the individual memory cells is elevationally along individual of the charge-blocking regions. The insulative charge-passage material (e.g., a band gap-engineered structure having nitrogen-containing material [e.g., silicon nitride] sandwiched between two insulator oxides [e.g., silicon dioxide]) is laterally between the channel material and the storage material.

FIGS. 1-6 show one embodiment wherein charge-blocking material 30, storage material 32, and charge-passage material 34 have been formed in individual channel openings 25 elevationally along insulative tiers 20 and conductive tiers 22. Transistor materials 30, 32, and 34 (e.g., memory-cell materials) may be formed by, for example, deposition of respective thin layers thereof over stack 18 and within individual channel openings 25 followed by planarizing such back at least to a top surface of stack 18 as shown.

Channel material 36 has also been formed in channel openings 25 elevationally along insulative tiers 20 and conductive tiers 22 and comprise individual operative channel-material strings 53 in one embodiment having memory-cell materials (e.g., 30, 32, and 34) there-along and with material 24 in insulative tiers 20 being horizontally-between immediately-adjacent channel-material strings 53. Materials 30, 32, 34, and 36 are collectively shown as and only designated as material 37 in some figures due to scale. Example channel materials 36 include appropriately-doped crystalline semiconductor material, such as one or more silicon, germanium, and so-called III/V semiconductor materials (e.g., GaAs, InP, GaP, and GaN). Example thickness for each of materials 30, 32, 34, and 36 is 25 to 100 Angstroms. Punch etching may be conducted as shown to remove materials 30, 32, and 34 from the bases of channel openings 25 to expose conductor tier 16 such that channel material 36 is directly against conductor material 17 of conductor tier 16. Such punch etching may occur separately with respect to each of materials 30, 32, and 34 (as shown) or may occur collectively with respect to all after deposition of material 34 (not shown). Alternately, and by way of example only, no punch etching may be conducted and channel material 36 may be directly electrically coupled to conductor material 17 of conductor tier 16 by a separate conductive interconnect (not shown). Channel openings 25 are shown as comprising a radially-central solid dielectric material 38 (e.g., spin-on-dielectric, silicon dioxide, and/or silicon nitride). Alternately, and by way of example only, the radially-central portion within channel openings 25 may include void space(s) (not shown) and/or be devoid of solid material (not shown).

5

Example conductive tiers **22** comprise conducting metal material **48** that is part of individual horizontally-elongated conductive lines **29** (e.g., wordlines) that are also part of elevationally-extending strings **49** of individual transistors and/or memory cells **56**. Example metal materials **48** include W, Ru, Co, Ti, Cu, Ni, Al, and conductive nitrides thereof (stoichiometric or non-stoichiometric). Approximate locations of transistors and/or memory cells **56** are indicated with a bracket or with dashed outlines, with transistors and/or memory cells **56** being essentially ring-like or annular in the depicted example. Alternately, transistors and/or memory cells **56** may not be completely encircling relative to individual channel openings **25** such that each channel opening **25** may have two or more elevationally-extending strings **49** (e.g., multiple transistors and/or memory cells about individual channel openings in individual conductive tiers with perhaps multiple wordlines per channel opening in individual conductive tiers, and not shown). Conducting material **48** may be considered as having terminal ends **50** corresponding to control-gate regions **52** of individual transistors and/or memory cells **56**. Control-gate regions **52** in the depicted embodiment comprise individual portions of individual conductive lines **29**. Materials **30**, **32**, and **34** may be considered as comprising a structure **65** that is laterally between control-gate region **52** and channel material **36**.

In the depicted embodiment, charge-passage material **34** is in conductive tiers **22** laterally-outward of channel-material strings **53** (e.g., laterally-outward along a direction orthogonal to direction **55** in FIGS. **2** and **4-6**) and storage material **34** is in conductive tiers **22** laterally-outward of charge-passage material **34**.

A charge-blocking region (e.g., charge-blocking material **30**) is between storage material **32** and individual control-gate regions **52**. A charge block may have the following functions in a memory cell: In a program mode, the charge block may prevent charge carriers from passing out of the storage material (e.g., floating-gate material, charge-trapping material, etc.) toward the control gate, and in an erase mode the charge block may prevent charge carriers from flowing into the storage material from the control gate. Accordingly, a charge block may function to block charge migration between the control-gate region and the storage material of individual memory cells. An example charge-blocking region as shown comprises insulator material **30**. By way of further examples, a charge-blocking region may comprise a laterally (e.g., radially) outer portion of the storage material (e.g., material **32**) where such storage material is insulative (e.g., in the absence of any different-composition material between an insulative storage material **32** and conducting material **48**). Regardless, as an additional example, an interface of a storage material and conductive material of a control gate may be sufficient to function as a charge-blocking region in the absence of any separate-composition-insulator material **30**. Further, an interface of conducting material **48** with material **30** (when present) in combination with insulator material **30** may together function as a charge-blocking region, and as alternately or additionally may a laterally-outer region of an insulative storage material (e.g., a silicon nitride material **32**). An example material **30** is one or more of silicon hafnium oxide and silicon dioxide.

At least one of AlOq, ZrOq, and HfOq **70** is in conductive tiers **22** laterally-outward of, in one embodiment directly against, storage material **32** (e.g., where “q” is 0.7 to 2.3) and may comprise charge-blocking material, for example in combination with charge-blocking material **30** when present and, if so, regardless of position of materials **70** and **30**

6

relative one another, yet ideally as shown. Charge-blocking material **30** may not be present (not shown). In one embodiment and as shown, at least one of AlOq, ZrOq, and HfOq **70** is directly above and directly below metal material **48** of individual horizontally-elongated conductive lines **29**. Regardless, in one embodiment, the at least one of AlOq, ZrOq, and HfOq is at least two of AlOq, ZrOq, and HfOq.

At least one of (a) and (b) is in conductive tiers **22** laterally-outward of, in one embodiment directly against, at least one of AlOq, ZrOq, and HfOq **70**, where,

(a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and

(b): MoM_z , where “M” is at least one of W, a Group 7 metal (i.e., herein any of Mn, Tc, Re, Bh, Pm, and Np), and a Group 8 metal (i.e., herein any of Fe, Ru, Os, Hs, Sm, Pu), with “z” being greater than 0 and less than 1.0.

The (a) and/or the (b) is designated with numeral **75** and materials **70** and **75** are collectively shown with a thick line in FIGS. **1** and **2** for clarity and due to scale. The (a) and/or the (b) may be deposited using any technique, for example by chemical vapor deposition or atomic layer deposition (MoO_2Cl_2 being an example precursor for each), or by physical vapor deposition.

Metal material **48** of individual horizontally-elongated conductive lines **29** in conductive tiers **22** is laterally-outward of the at least one of the (a) and the (b) **75**. In one embodiment and as shown, the at least one of the (a) and the (b) **75** is directly above and directly below metal material **48** of individual horizontally-elongated conductive lines **29**. In one embodiment, the at least one of the (a) and the (b) **75** is directly against at least one of AlOq, ZrOq, and HfOq **70** and in one embodiment the at least one of the (a) and the (b) **75** is directly against metal material **48**.

In one embodiment, the at least one of the (a) and the (b) comprises the (a), in one such embodiment where “x” is zero and in one such latter embodiment where “y” is not zero. In one embodiment, the at least one of the (a) and the (b) is the (a) and “y” is zero, and in one such embodiment “x” is not zero. In one embodiment, the at least one of the (a) and the (b) is (a) and each of “x” and “y” is zero. In one embodiment, the at least one of the (a) and the (b) is the (a) and “x” is no greater than 1.0 and in one such embodiment where “x” is 1.0 and in one such embodiment where “y” is zero.

In one embodiment, the at least one of the (a) and the (b) is the (a) and “x” is greater than 1.0, and in one such embodiment “y” is zero. In one embodiment, the at least one of the (a) and the (b) is the (a) and “y” is no greater than 1.0, and in one such embodiment where “y” is 1.0 and in one such embodiment where “x” is zero. In one embodiment, the at least one of the (a) and the (b) is the (a) and “y” is greater than 1.0, and in one such embodiment “x” is zero.

In one embodiment, the at least one of the (a) and the (b) comprises the (b). In one such embodiment, “M” is only one of W, a Group 7 metal, and a Group 8 metal and alternately in one such embodiment “M” is more than one of W, a Group 7 metal, and a Group 8 metal. In one embodiment, at least one of the (a) and the (b) comprises the (b) and “z” is from 0.5 to 0.25. In one embodiment, at least one of the (a) and the (b) comprises the (b) and “M” comprises W. In one embodiment, at least one of the (a) and the (b) comprises the (b) and “M” comprises a Group 7 metal. In one embodiment, the at least one of the (a) and the (b) comprises the (b) and “M” comprises a Group 8 metal.

Memory cells **56** in individual conductive tiers **22** individually comprise channel material **36** of individual channel-material strings **53**, storage material **32**, at least one of

AlOq, ZrOq, and HfOq **70**, the at least one of the (a) and the (b) **75**, and metal material **48**.

In one embodiment, memory circuitry comprising strings (e.g., **49**) of memory cells (e.g., **56**) comprises a vertical stack (e.g., **18**) comprising alternating insulative tiers (e.g., **20**) and conductive tiers (e.g., **22**). Channel-material strings (e.g., **53**) extend through the insulative tiers and the conductive tiers. Charge-passage material (e.g., **34**) is in the conductive tiers laterally-outward of the channel-material strings. Storage material (e.g., **32**) is in the conductive tiers laterally-outward of the charge-passage material. At least one of AlOq, ZrOq, and HfOq (e.g., **70**) is in the conductive tiers laterally-outward of the storage material. At least one of (a) and (b) (e.g., **75**) is in the conductive tiers laterally-outward of, in one embodiment directly against, the at least one of AlOq, ZrOq, and HfOq, where,

(a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and

(b): MoM_z , where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0.

Metal material (e.g., **48**) is in the conductive tiers laterally-outward of the at least one of the (a) and the (b). Memory cells (e.g., **56**) are in individual of the conductive tiers and comprise the channel material of individual of the channel-material strings, the storage material, the at least one of AlOq, ZrOq, and HfOq, the at least one of the (a) and the (b), and the metal material. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

Embodiments of the invention include one or more transistors regardless of whether comprising a memory cell. Constructions **56** as shown individually comprise such a transistor where, for example, conductive material **17** comprises one source/drain region [e.g., a source region] of the transistor(s). Another of the source/drain regions [e.g., a drain region] may be atop or part of an upper portion of stack **18** [not numerically designated]. In one embodiment and as shown with respect to FIGS. 1-6, such transistor(s)/memory cell(s) comprising construction **56** is a vertical transistor. Alternately, as another example, the transistor may be horizontal, for example as shown in FIG. 7 with respect to a transistor **85** of a construction **10a**. Like numerals from the above-described embodiments have been used where appropriate, with some construction differences being indicated with the suffix “a” or with different numerals.

Transistor **85** comprises channel material **36a** between a pair of source/drain regions **86** and **88**. Charge-passage material **34a** is adjacent, in one embodiment directly against, channel material **36a**. Storage material **32a** is adjacent, in one embodiment directly against, charge-passage material **34a**. At least one of AlOq, ZrOq, and HfOq **70a** is adjacent, in one embodiment directly against, storage material **32a**. At least one of (a) and (b) **75** is adjacent, in one embodiment directly against, at least one of AlOq, ZrOq, and HfOq **70a**, where:

(a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and

(b): MoM_z , where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0.

A control gate (e.g., as a portion of line **29a**) comprising metal material **48** is adjacent, in one embodiment directly against, the at least one of the (a) and the (b) **75**. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

The above processing(s) or construction(s) may be considered as being relative to an array of components formed as or within a single stack or single deck of such components above or as part of an underlying base substrate (albeit, the single stack/deck may have multiple tiers). Control and/or other peripheral circuitry for operating or accessing such components within an array may also be formed anywhere as part of the finished construction, and in some embodiments may be under the array (e.g., CMOS under-array). Regardless, one or more additional such stack(s)/deck(s) may be provided or fabricated above and/or below that shown in the figures or described above. Further, the array(s) of components may be the same or different relative one another in different stacks/decks and different stacks/decks may be of the same thickness or of different thicknesses relative one another. Intervening structure may be provided between immediately-vertically-adjacent stacks/decks (e.g., additional circuitry and/or dielectric layers). Also, different stacks/decks may be electrically coupled relative one another. The multiple stacks/decks may be fabricated separately and sequentially (e.g., one atop another), or two or more stacks/decks may be fabricated at essentially the same time.

The assemblies and structures discussed above may be used in integrated circuits/circuitry and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multichip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

In this document unless otherwise indicated, “elevational”, “higher”, “upper”, “lower”, “top”, “atop”, “bottom”, “above”, “below”, “under”, “beneath”, “up”, and “down” are generally with reference to the vertical direction. “Horizontal” refers to a general direction (i.e., within 10 degrees) along a primary substrate surface and may be relative to which the substrate is processed during fabrication, and vertical is a direction generally orthogonal thereto. Reference to “exactly horizontal” is the direction along the primary substrate surface (i.e., no degrees there-from) and may be relative to which the substrate is processed during fabrication. Further, “vertical” and “horizontal” as used herein are generally perpendicular directions relative one another and independent of orientation of the substrate in three-dimensional space. Additionally, “elevationally-extending” and “extend(ing) elevationally” refer to a direction that is angled away by at least 450 from exactly horizontal. Further, “extend(ing) elevationally”, “elevationally-extending”, “extend(ing) horizontally”, “horizontally-extending” and the like with respect to a field effect transistor are with reference to orientation of the transistor’s channel length along which current flows in operation between the source/drain regions. For bipolar junction transistors, “extend(ing) elevationally”, “elevationally-extending”, “extend(ing) horizontally”, “horizontally-extending” and the like, are with reference to orientation of the base length along which current flows in operation between the emitter and collector. In some embodiments, any component, feature, and/or region that extends elevationally extends vertically or within 10° of vertical.

Further, “directly above”, “directly below”, and “directly under” require at least some lateral overlap (i.e., horizon-

tally) of two stated regions/materials/components relative one another. Also, use of “above” not preceded by “directly” only requires that some portion of the stated region/material/component that is above the other be elevationally outward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components). Analogously, use of “below” and “under” not preceded by “directly” only requires that some portion of the stated region/material/component that is below/under the other be elevationally inward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components).

Any of the materials, regions, and structures described herein may be homogenous or non-homogenous, and regardless may be continuous or discontinuous over any material which such overlie. Where one or more example composition(s) is/are provided for any material, that material may comprise, consist essentially of, or consist of such one or more composition(s). Further, unless otherwise stated, each material may be formed using any suitable existing or future-developed technique, with atomic layer deposition, chemical vapor deposition, physical vapor deposition, epitaxial growth, diffusion doping, and ion implanting being examples.

Additionally, “thickness” by itself (no preceding directional adjective) is defined as the mean straight-line distance through a given material or region perpendicularly from a closest surface of an immediately-adjacent material of different composition or of an immediately-adjacent region. Additionally, the various materials or regions described herein may be of substantially constant thickness or of variable thicknesses. If of variable thickness, thickness refers to average thickness unless otherwise indicated, and such material or region will have some minimum thickness and some maximum thickness due to the thickness being variable. As used herein, “different composition” only requires those portions of two stated materials or regions that may be directly against one another to be chemically and/or physically different, for example if such materials or regions are not homogenous. If the two stated materials or regions are not directly against one another, “different composition” only requires that those portions of the two stated materials or regions that are closest to one another be chemically and/or physically different if such materials or regions are not homogenous. In this document, a material, region, or structure is “directly against” another when there is at least some physical touching contact of the stated materials, regions, or structures relative one another. In contrast, “over”, “on”, “adjacent”, “along”, and “against” not preceded by “directly” encompass “directly against” as well as construction where intervening material(s), region(s), or structure(s) result(s) in no physical touching contact of the stated materials, regions, or structures relative one another.

Herein, regions-materials-components are “electrically coupled” relative one another if in normal operation electric current is capable of continuously flowing from one to the other and does so predominately by movement of subatomic positive and/or negative charges when such are sufficiently generated. Another electronic component may be between and electrically coupled to the regions-materials-components. In contrast, when regions-materials-components are referred to as being “directly electrically coupled”, no intervening electronic component (e.g., no diode, transistor, resistor, transducer, switch, fuse, etc.) is between the directly electrically coupled regions-materials-components.

Any use of “row” and “column” in this document is for convenience in distinguishing one series or orientation of features from another series or orientation of features and along which components have been or may be formed.

“Row” and “column” are used synonymously with respect to any series of regions, components, and/or features independent of function. Regardless, the rows may be straight and/or curved and/or parallel and/or not parallel relative one another, as may be the columns. Further, the rows and columns may intersect relative one another at 90° or at one or more other angles (i.e., other than the straight angle).

The composition of any of the conductive/conductor/conducting materials herein may be metal material and/or conductively-doped semiconductive/semiconductor/semi-conducting material. “Metal material” is any one or combination of an elemental metal, any mixture or alloy of two or more elemental metals, and any one or more conductive metal compound(s).

Herein, any use of “selective” as to etch, etching, removing, removal, depositing, forming, and/or formation is such an act of one stated material relative to another stated material(s) so acted upon at a rate of at least 2:1 by volume. Further, any use of selectively depositing, selectively growing, or selectively forming is depositing, growing, or forming one material relative to another stated material or materials at a rate of at least 2:1 by volume for at least the first 75 Angstroms of depositing, growing, or forming.

Unless otherwise indicated, use of “or” herein encompasses either and both.

CONCLUSION

In some embodiments, memory circuitry comprising strings of memory cells comprises a vertical stack comprising alternating insulative tiers and conductive tiers. Channel-material strings extend through the insulative tiers and the conductive tiers. Charge-passage material is in the conductive tiers laterally-outward of the channel-material strings. Storage material is in the conductive tiers laterally-outward of the charge-passage material. At least one of AlOq, ZrOq, and HfOq is in the conductive tiers laterally-outward of the storage material. At least one of (a) and (b) is in the conductive tiers laterally-outward of the at least one of AlOq, ZrOq, and HfOq, where, (a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and (b): MoM_z , where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0. Metal material is in the conductive tiers laterally-outward of the at least one of the (a) and the (b). Memory cells are in individual of the conductive tiers. The memory cells individually comprise the channel material of individual of the channel-material strings, the storage material, the at least one of AlOq, ZrOq, and HfOq, the at least one of the (a) and the (b), and the metal material.

In some embodiments, memory circuitry comprising strings of memory cells comprises laterally-spaced memory blocks individually comprising a vertical stack comprising alternating insulative tiers and conductive tiers. Channel-material strings extend through the insulative tiers and the conductive tiers. The conductive tiers individually comprise a horizontally- elongated conductive line comprising metal material. Charge- passage material is in the conductive tiers laterally-outward of the channel-material strings. Storage material is in the conductive tiers laterally-outward of the charge-passage material. At least one of AlOq, ZrOq, and HfOq is in the conductive tiers laterally-outward of the storage material. At least one of (a) and (b) is in the

11

conductive tiers laterally-outward of the at least one of AlOq, ZrOq, and HfOq, where, (a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and (b): MoM_z , where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0. The metal material of individual of the horizontally-elongated conductive lines in the conductive tiers is laterally-outward of the at least one of the (a) and the (b). The at least one of AlOq, ZrOq, and HfOq and the at least one of the (a) and the (b) is directly above and directly below the metal material. Memory cells are in individual of the conductive tiers. The memory cells individually comprise the channel material of individual of the channel-material strings, the storage material, the at least one of AlOq, ZrOq, and HfOq, the at least one of the (a) and the (b), and the metal material.

In some embodiments, a transistor comprises channel material between a pair of source/drain regions. Charge-passage material is adjacent the channel material. Storage material is adjacent the charge-passage material. At least one of AlOq, ZrOq, and HfOq is adjacent the storage material. At least one of (a) and (b) is adjacent the at least one of AlOq, ZrOq, and HfOq, where, (a): MoO_xN_y , where each of “x” and “y” is from 0 to 4.0; and (b): MoM_z , where “M” is at least one of W, a Group 7 metal, and a Group 8 metal; “z” being greater than 0 and less than 1.0. Control gate comprising metal material is adjacent the at least one of the (a) and the (b).

In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

The invention claimed is:

1. Memory circuitry comprising strings of memory cells, comprising:

- a vertical stack comprising alternating insulative tiers and conductive tiers, channel-material strings extending through the insulative tiers and the conductive tiers;
- a charge-passage material in the conductive tiers laterally-outward of the channel-material strings;
- a storage material in the conductive tiers laterally-outward of the charge-passage material, the storage material extending an entirety of a vertical height of each of the conductive tiers;
- a metal oxide consisting of at least one of AlOq, ZrOq, and HfOq in the conductive tiers laterally-outward of the storage material, the metal oxide having upper and lower surfaces in direct physical contact with an insulative material of the insulative tiers;
- MoM_z in the conductive tiers laterally-outward of the at least one of AlOq, ZrOq, and HfOq, where “M” is at least one of Tc, Bh, Fe, Os, and Hs; “z” being greater than 0 and less than 1.0;
- a metal material in the conductive tiers laterally-outward of the MoM_z ; and
- memory cells in individual of the conductive tiers, the memory cells individually comprising:
 - a channel material of individual of the channel-material strings;
 - the storage material;
 - the metal oxide;
 - the MoM_z ; and
 - the metal material.

12

2. The memory circuitry of claim 1 wherein the M is Tc.

3. The memory circuitry of claim 1 wherein the M is Bh.

4. The memory circuitry of claim 1 wherein the M is Fe.

5. The memory circuitry of claim 1 wherein the M is Os.

6. The memory circuitry of claim 1 wherein the M is Hs.

7. The memory circuitry of claim 1 wherein “M” is only one of Tc, Bh, Fe, Os, and Hs.

8. The memory circuitry of claim 1 wherein “M” is more than one of Tc, Bh, Fe, Os, and Hs.

9. The memory circuitry of claim 1 wherein “z” is from 0.5 to 0.25.

10. The memory circuitry of claim 1, further comprising a charge-blocking material between and in direct physical contact with the metal oxide and the storage material.

11. The memory circuitry of claim 1 wherein the storage material is in direct physical contact with the metal oxide.

12. Memory circuitry comprising strings of memory cells, comprising:

laterally-spaced memory blocks individually comprising a vertical stack comprising alternating insulative tiers and conductive tiers, channel-material strings extending through the insulative tiers and the conductive tiers, the conductive tiers individually comprising a horizontally-elongated conductive line comprising a metal material;

a charge-passage material in the conductive tiers laterally-outward of the channel-material strings;

a storage material in the conductive tiers laterally-outward of the charge-passage material;

a metal oxide comprising at least one of AlOq, ZrOq, and HfOq in the conductive tiers laterally-outward of the storage material, the metal oxide being in direct physical contact with an insulative material of the insulative tiers and extending vertically along the insulative tiers; MoM_z in the conductive tiers laterally-outward of the at least one of AlOq, ZrOq, and HfOq, where “M” is at least one of Tc, Bh, Fe, Os, and Hs; “z” being greater than 0 and less than 1.0;

the metal material of individual of the horizontally-elongated conductive lines in the conductive tiers being laterally-outward of the MoM_z , the at least one of AlOq, ZrOq, and HfOq and the MoM_z being directly above and directly below the metal material; and

memory cells in individual of the conductive tiers, the memory cells individually comprising:

a channel material of individual of the channel-material strings;

the storage material;

the metal oxide;

the MoM_z ; and

the metal material.

13. The memory circuitry of claim 12 wherein the M is Tc.

14. The memory circuitry of claim 12 wherein the M is Bh.

15. The memory circuitry of claim 12 wherein the M is Fe.

16. The memory circuitry of claim 12 wherein the M is Os.

17. The memory circuitry of claim 12 wherein the M is Hs.

18. The memory circuitry of claim 12 wherein “M” is only one of Tc, Bh, Fe, Os, and Hs.

19. The memory circuitry of claim 12 wherein “M” is more than one of Tc, Bh, Fe, Os, and Hs.

13

14

20. The memory circuitry of claim **12** wherein “z” is from 0.5 to 0.25.

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